Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A wave travels 9 m in 0.3 s at frequency 50 Hz. Find speed and wavelength.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: λ = v/f; compare opening width with λ. Spreading is pronounced when the opening width is comparable to wavelength. Label the opening and wavefronts.

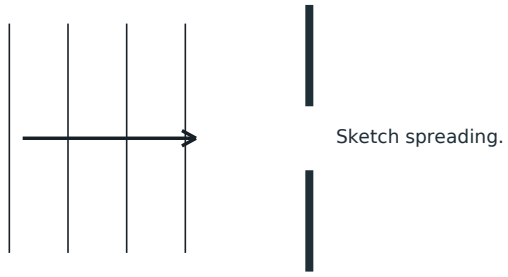
Worked reasoning: A wave with wavelength 1 m meets openings 1 m and 10 m wide. Spreading relative to opening size is more noticeable for the 1 m opening.

Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.

Marzano check - circle: ① ② ③

03 Represent and interpret

Sketch wavefronts after a narrow opening about one wavelength wide. Compare with a much wider opening.



Marzano check - circle: ① ② ③

04 Solve and justify

A sound wave has frequency 340 Hz and speed 340 m/s. Findλ and compare a .8 m doorway with a .0000005 m visible-light wavelength. Why does heard-around-corner behavior differ?

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Marzano check - circle: 1 2 3

06 Choose a method

For a fixed ideal-gas particle mass, average translational kinetic energy becomes 16 times as large. Compare rms speed.

Marzano check - circle: 1 2 3

07 Exit check

At fixed wave speed, frequency doubles. Explain how$\lambda$and typical diffraction through a fixed opening change.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Sound speed is 340 m/s. Compare wavelengths at 170 and 1700 Hz with a 1 m doorway. Explain which should diffract more noticeably.

Marzano check - circle: 1 2 3